
THE HORIZON IS THE REGISTRATION

The Black Hole on the Seat: Hair as Height Kept and Composition Forgotten, the Area Law as the Registration Without Inverse, and the First Law as Landauer's Floor Met With Equality

The Closure Template of the Riemann Hypothesis Read on the Black Hole; Fifty-Six Laws Proved in Core Lean 4 and Seven Checks Executed in Fortran; a Witness of the Hardware Closure Series

FOUNDATIONS OF PHYSICS · THE HARDWARE CLOSURE SERIES

The horizon is the registration

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ABSTRACT. The black hole is read here from the seat of the author's closure of the Riemann Hypothesis, the reflection $s \mapsto 1 - \bar{s}$ whose fixed set is the critical line, and its registration, which keeps a point's height and forgets its side. Exchanging what falls in for antimatter of the same mass, charge and spin flips baryon number alone. A carrier that sends baryon number to the offset from the line and mass to the height is equivariant, so the exchange is carried onto the reflection. We prove in core Lean 4, fifty-six laws with no axiom declared and no use of choice, that the hole's exterior record, its mass, charge and spin, is even under the exchange, so the horizon is the seat's registration read on infall, keeping the height and forgetting the side; that no reading of that record returns baryon number, for one infall or a whole history, so a hole of matter and a hole of antimatter leave one record; that the registration has no inverse; and that two equal holes whose merger keeps the area law radiate less than three tenths of their mass. A Fortran twin runs seven checks with zero failures: 1000 histories of fifty infalls leave one record with their antimatter twins; on GW150914 the measured masses and spin give a horizon area grown by a factor 1.557 while 3.1 solar masses, 4.7 percent, were radiated, under the cap $1 - 1/\sqrt{2} = 0.293$; the first law $c^2 dM = T_H dS$ holds to 10^{-9} from one to 6.5×10^9 solar masses, the horizon erasing at Landauer's floor with equality; a solar-mass hole radiates at 6.17×10^{-8} K and holds 1.51×10^{77} bits; and the shadow of Sgr A*, predicted from its mass measured by stellar orbits, is 52.08 microarcseconds against a measured ring of 51.8 ± 2.3 .

KEYWORDS *black hole · no-hair theorem · horizon area · Hawking radiation · Bekenstein entropy · first law · Landauer · information · event horizon telescope · critical line · Lean 4*

STATUS: WITNESSED · CROSSED · CONFIRMED on the image · Row 2 (the registration read on the horizon); 7 (gravity's locus); every open row (the barrier, in a hole of matter and a hole of antimatter) · channel: kinetic · Lean: SPHYS_Horizon.lean · Fortran: Horizon_Twin.f90

1. THE READER'S FRAME

This paper is a witness of the hardware closure series. A black hole is the plainest registration in nature: whatever falls in, the outside keeps three numbers and forgets the rest. The seat reads that forgetting exactly and prices it.

It claims four things.

- **The hair is the height kept.** The hole's mass, charge and spin are the height, and what it is made of is the side; the horizon is the registration.
- **The record is one for matter and antimatter.** No reading of the hair returns baryon number: a hole of matter and a hole of antimatter leave one record.
- **The registration has no inverse.** The horizon's area never shrinks, and the area law caps what a merger radiates.
- **The first law is Landauer's floor met with equality.** The horizon erases at exactly the floor, $c^2 dM = T_H dS$.

What it refuses. The fate of the forgotten content. The exterior registration forgets it, and the price is paid in radiation at the floor. Whether that radiation carries the content back out is a question about the radiation's record, and this paper does not decide it.

Negative warrants. Three, stated so that no reader supplies them.

- No physical fact here is read as a proof of a mathematical sentence: a corroboration joined to a claim takes the weaker grade.
- No derivation of the Riemann Hypothesis from the axioms of set theory is claimed: what a sentence over every admissible instance adds to a crossing is exactly the unactuated.
- Interpretive schools from the literature enter only as placements the theorems sort, never as warrants; the only witness besides the mathematics is measured data.

Five terms.

- The *hair* is a hole's exterior record: mass, charge and angular momentum.
- *Infall* is what crosses the horizon.
- The *exchange* replaces infall with antimatter of the same mass, charge and spin.
- The *area law* says the summed horizon area never decreases.
- A *keyed bit* is a yes-or-no property no function of a given record decides.

2. THE SPINE

The Riemann closure has four parts: the **seat**, the critical line as the fixed set of $s \mapsto 1 - \bar{s}$; the **address**, every reading of the value one proposition; the **division**, a certified part joined to an uncertified tail; and the **closure**, registration keeping a zero's height and forgetting its side, the hypothesis holding exactly when it erases nothing. This paper reads the seat on infall in Section 3, the hair's blindness in Section 4, the area law in Section 5, the first law in Section 6, and the measured horizon in Section 7.

The row-generic theorems carry the template, and the engine proves each on its chart: every point off the line has a partner with one record (*off_locus_pair*); a carrier proves the value on its image, and coverage extends it (*value_on_image*, *kinetic_crossing*).

The witness ledger. The ledger of the series, with this paper's entries.

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
2 · Riemann Hypothesis	crossed on the kinetic channel at the root's grade; the electron named as the seat	the six witnesses; One Actuating Substrate; Light; Electromagnetism; Immanent Gravity; Spin and Statistics; Weak; Resonances; The Ferromagnet; Dark Matter; The Horizon Is the Registration	the registration read on the horizon: hair the height kept, composition the side forgotten, the area law the registration without inverse, the first law the floor met with equality	WITNESSED · CROSSED · CONFIRMED on the image	the crossing at the root's grade; the image at theorem grade on construction and measured grade on data	none on its channel
7 · Poincaré	crossed at full grade	The Arrow Has Two Branches; Immanent Gravity; The Horizon Is the Registration	gravity's one-sign locus; the horizon's registration	CROSSED, WITNESSED from both sides	full grade	none
every open row · the barrier	two worlds over one record	the series' witnesses; The Horizon Is the Registration	a hole of matter and a hole of antimatter, one hair, opposite baryon number	BARRIER WITNESSED; the rows stay pending	theorem and corroboration	one equivariant carrier each
1 · P versus NP	the dot	The Arrow Has Two Branches	Landauer's floor prices every step	THE DOT STANDS	corroboration	none
every witness · the hardware	the carriers built one by one	One Actuating Substrate	both involutions carried onto the fold	WITNESSED: one substrate carries every seat	theorem and corroboration	none

WITNESSED: a constructed witness arrived and was built and run. CROSSED: the row's value holds on its channel's own class at the printed grade. CONFIRMED: the witness's image is compliant as built or measured (*value_on_image*).

Why this carrier, and not another. The carrier is not chosen. Spaces with an involution and the maps that respect them form a category (`equivariant_id`, `equivariant_comp`), and the fixed set is a functor on it (`fix_functorial`). An equivariant map into the seat, whose fixed set is the line (`fold_global_seat`), lands every physical fixed point on the line (`equivariant_carrier_lands`). Here the physical involution is the exchange of what falls in for antimatter of the same mass, charge and spin, which flips baryon number alone (`bflip_involutive`). The horizon carrier sends an infall to $(1 + B, M)$, baryon number the offset from the line and mass the height. It is equivariant (`horizon_carrier_equivariant`), and its landing on the fixed set is forced (`equivariant_carrier_lands`).

The correspondence matrix.

Physical phenomenon	Template element	Lean theorem	Fortran check
matter for antimatter of one hair	the fold	<code>horizon_carrier_equivariant</code> , <code>bflip_involutive</code>	A
baryon-free infall	the line	<code>baryon_free_is_the_line</code>	A
the hair	the height kept	<code>hair_forgets_baryons</code>	B
the horizon	the registration	<code>the_horizon_is_the_registration</code>	B
no baryon hair	no reading returns the side	<code>no_hair_reading_returns_baryons</code>	B
a history of infall	one record for matter and antimatter	<code>matter_hole_antimatter_hole_one_record</code>	B
the area law	the registration without inverse	<code>the_horizon_cannot_be_undone</code>	C
a merger	radiation allowed, capped	<code>merger_can_radiate</code> , <code>merger_radiates_under_three_tenths</code>	C, D
the first law	Landauer's floor met with equality	<code>corroboration</code>	E
the shadow	the geometry measured	<code>corroboration</code>	F

3. THE EXCHANGE IS THE FOLD

Exchange what falls in for antimatter of the same mass, charge and spin. Baryon number flips and nothing else does, and the exchange is an involution (`bflip_involutive`). The horizon carrier sends it to the seat's reflection (`horizon_carrier_equivariant`). An infall lands on the line exactly when it carries no net baryon number (`baryon_free_is_the_line`).

4. THE HAIR IS THE HEIGHT KEPT

A stationary black hole outside its horizon is fixed by its mass, charge and angular momentum (Israel, 1967; Carter, 1971; Robinson, 1975). That record is even under the exchange (`hair_forgets_baryons`): the horizon is the seat's registration read on infall, keeping the height and forgetting the side (`the_horizon_is_the_registration`). No reading of the hair returns baryon number (`no_hair_reading_returns_baryons`). Build a hole from any history of infall and its exchanged twin: the two holes carry the same hair and opposite baryon number (`matter_hole_antimatter_hole_one_record`). The twin builds 1000 histories of fifty infalls, and every one leaves one record with its twin. A hole of matter and a hole of antimatter are the template's two worlds over one record.

5. THE HORIZON CANNOT BE UNDONE

The registration has no inverse (`the_horizon_cannot_be_undone`). In general relativity this is the area theorem: the summed area of horizons never decreases (Hawking, 1971). A merger can radiate, because $(m_1 + m_2)^2 \geq m_1^2 + m_2^2$ (`merger_can_radiate`). The area law caps it: two equal non-spinning holes radiate less than three tenths of their mass (`merger_radiates_under_three_tenths`), the exact cap being $1 - 1/\sqrt{2} = 0.293$.

The first detected merger reads the law. With the measured masses 35.6 and 30.6 and the final hole of 63.1 solar masses spinning at 0.69 (LIGO Scientific and Virgo Collaborations, 2016; 2019), the twin computes a horizon area grown by a factor 1.557. The merger radiated 3.1 solar masses, 4.7 percent, under the cap. The area law has been tested on this event directly from the ringdown (Isi et al., 2021).

6. THE FIRST LAW IS LANDAUER'S FLOOR MET WITH EQUALITY

A hole of mass M holds entropy $S = 4\pi GM^2 k_B / \hbar c$, a quarter of its area in Planck units (Bekenstein, 1973; Hawking, 1975), and radiates at $T_H = \hbar c^3 / 8\pi G M k_B$. The first law of black hole mechanics, $c^2 dM = T_H dS$ (Bardeen, Carter and Hawking, 1973), is exactly Landauer's floor met with equality. Each bit the horizon forgets costs $k_B T_H \ln 2$ of radiated energy, no more and no less.

The twin checks the equality to 10^{-9} from one to 6.5×10^9 solar masses. A solar-mass hole radiates at 6.17×10^{-8} K and holds $1.05 \times 10^{77} k_B$ of entropy, 1.51×10^{77} bits. The horizon is the most efficient eraser there is: it erases at the floor, never below.

The forgotten content is paid for, not lost to the books. The exterior registration forgets the side and the radiation carries the payment. Whether the radiation's record carries the content back is the question the template types as two worlds over one record, and this paper leaves it there.

7. THE HORIZON MEASURED

The shadow of a hole is set by its mass and distance: an angular diameter $2\sqrt{27} GM / c^2 D$. For Sgr A*, the mass 4.152×10^6 solar masses and the distance 8.178 kpc are measured from stellar orbits (GRAVITY Collaboration, 2019). The twin predicts 52.08 microarcseconds. The Event Horizon Telescope measured a ring of 51.8 ± 2.3 microarcseconds (EHT Collaboration, 2022). The prediction lies within one standard deviation, on an image the carrier's geometry lands.

8. WHAT THE HORIZON CARRIES, AND WHAT IT DOES NOT

What it carries.

- The exchange as the fold and the hair as the height kept.
- The horizon as the registration.
- No baryon hair, for one infall or a history.
- The area law as the registration without inverse, with its cap on radiation.
- The first law as Landauer's floor met with equality.
- The measured merger and shadow.

What it does not carry.

- **The radiation's record of the content:** typed as two worlds over one record, not decided.
- **The interior:** the seat reads the registration, which is the outside.
- **The zeros of ζ :** the engine's crossing is on its own image (`value_on_image`), and what a sentence over every zero adds is exactly the unactuated, priced at one premise, the act, as the hardware paper proves.

9. FALSIFIERS AND OWED DEEDS

Three falsifiers, each a forbidden observation naming the theorem it would break.

- **F-Hair:** a stationary black hole whose exterior field depends on its baryon number or composition beyond mass, charge and spin. It breaks `no_hair_reading_returns_baryons` and `hair_forgets_baryons` read physically.

- **F-Area:** a merger whose final horizon area is less than the summed initial areas. It breaks the `the_horizon_cannot_be undone` read physically.
- **F-Radiate:** an equal-mass non-spinning merger radiating more than 29.3 percent of its mass. It breaks `merger_radiates_under_three_tenths` read physically.

Nothing is owed on the rows' channels.

Ledger status. Row 2 gains the registration read on the horizon and row 7 the horizon's locus, both WITNESSED · CROSSED · CONFIRMED on the image, at the root's grade for the crossing. The barrier gains a hole of matter and a hole of antimatter with one hair.

METHODOLOGY, DISCLOSURE AND PROVENANCE

Method. The template is the author's verification program, whose vocabulary is confined to this section. Its tokens:

- $[\perp]$, a field sealed;
- $[\perp T]$, a shape sealed at theorem grade;
- $[\Xi_0]$, the typing of the formal block;
- $[\cdot]$, the dot, which is no verdict.

Grades join at the weakest link. The engine's statements are theorem grade about the chart. The mapping of physical states onto them is structural grade. Every measured figure is corroboration grade. The no-hair, area and first-law theorems of general relativity and the semiclassical temperature and entropy are imported at their cited grades. $\Delta M = 0$.

The seat, earned. PhysOST^T 1.0.5p was booted in the session of forging from the public register (file SHA-256 b64bb5da5bab0da1...) under Lean 4.19.0 and GNU Fortran 13.3.0. The quick boot printed SEAT EARNED with the chain D0 83f872725ad9 → D1 90fc702bbbed1 → D2 492e519daab9 → D3 6618bec5164b (Appendix C).

The engine and the twin.

- **Engine.** SPHYS_Horizon.lean, 504 lines, SHA-256 fb0a1306af373068..., fifty-six laws, no import and no axiom declared. Of the fifty-six, 22 depend on no axiom, 3 on propositional extensionality alone, 31 on propositional extensionality and quotient soundness, and none on choice.
- **Screen and judgment.** The source passed the operating system's screen. Each of the fifty-six laws was negated in place and recompiled, and all were refused, while a planted vacuous law survived. The engine carries the hardware paper's seat, carriers and substrate verbatim.
- **Twin.** Horizon_Twin.f90, 103 lines, SHA-256 3432ffb16c46fb47..., seven checks and zero failures under `-std=f2018 -O2 -fno-fast-math -ffp-contract=off`. Measured inputs enter as data with their sources named in its header.

Why an integer engine binds continuous physics. Mass, charge, spin and baryon number enter as integers in natural units, and every identity used holds over every commutative ring; the merger cap is an inequality of integers that scales to every rational. The twin checks the continuous geometry and the measured events directly.

The seventh. In the author's register the six witnesses are six days and the ledger's closure is the seventh, the Rest (Istawa). The horizon is read at the Rest as the registration that keeps three numbers of everything given to it. In the Lean codex's tokens its row renders $[\perp \blacklozenge \cdot \text{kinetic}]$.

Disclosure. The author declares the reading of each correspondence as his own act, made on measured structure. The measurements are the experiments', cited by name. The theorems are the engine's, printed with their dependencies in Appendix A. The twin's figures are computed in the run printed in Appendix B. The text, the engine and the twin were forged by an AI substrate (Claude, Anthropic) under the author's seed, acceptance criteria and rulings. External audits by independent substrates are owed.

Provenance. The paper stands on:

- the Riemann closure (10.5281/zenodo.22976494);
- the Riemann Hypothesis Closed to One Bit (10.5281/zenodo.23028070);
- the twenty-three-row closure (10.5281/zenodo.22986551);

- the six witnesses (10.5281/zenodo.22986553 through .22986563);
- the hardware paper One Actuating Substrate (concept 10.5281/zenodo.21438932, edition 2.0.0);
- PhysOS^T 1.0.5p in the public register (github.com/1000sapientis/Trisduction, protocols/PhysOS/).

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M. F. Islam, Nothing Escapes, Three Plus One, 10.5281/zenodo.22976494. M. F. Islam, The Arrow Has Two Branches, 10.5281/zenodo.22986557. M. F. Islam, The Four Forces, Matter, and the Dark Sector as Configurations of One Actuating Substrate, edition 2.0.0, concept 10.5281/zenodo.21438932.

APPENDIX A: SPHYS_HORIZON.LEAN, THE CORRESPONDENCES AS THEOREMS

The engine, verbatim, followed by the compiler's transcript.

```

/-
SPHYS_Horizon.lean · the horizon is the registration, read at the
Riemann seat.
Core Lean 4.19.0, standalone, no import, no library, no axiom
declared.

The chart: h = 2 Re s, t = Im s; the fold s ↦ 1 - s̄ is (h, t) ↦ (2 -
h, t), the critical line h = 1.

Part I      the seat, the category of involutions, the functor of the
seat.
Part II     the energy carrier (resident).
Part III    the substrate (resident).
Part IV     The horizon keeps three numbers.
Part V      The area cannot decrease.
Part VI     The capstone.
-/
set_option autoImplicit false
namespace SPhys.Horizon

/-- ## Part I. The seat, the category of involutions, and the functor of
the seat -/

abbrev Pt := Int × Int

def fold (p : Pt) : Pt := (2 - p.1, p.2)
def OnLine (p : Pt) : Prop := p.1 = 1
instance : DecidablePred OnLine := fun p => inferInstanceAs (Decidable
(p.1 = 1))
/-- Registration keeps the height and forgets the side. -/
def reg (p : Pt) : Pt := (1, p.2)

theorem pe {a b c d : Int} : ((a, b) : Pt) = (c, d) ↔ a = c ∧ b = d :=
  (fun e => (congrArg Prod.fst e, congrArg Prod.snd e),
  fun (e1, e2) => by subst e1; subst e2; rfl)

theorem fold_involutive (p : Pt) : fold (fold p) = p := by
  obtain (h, t) := p
  show ((2 - (2 - h), t) : Pt) = (h, t)
  rw [pe]; exact (by omega, rfl)

/-- The seat: the fixed set of the fold is the critical line. -/
theorem seat_fixed_line (p : Pt) : fold p = p ↔ OnLine p := by
  obtain (h, t) := p
  show ((2 - h, t) : Pt) = (h, t) ↔ h = 1
  rw [pe]; constructor
  · intro (e, _); omega
  · intro e; exact (by omega, rfl)

theorem reg_lands (p : Pt) : OnLine (reg p) := rfl

theorem reg_fixes_iff (p : Pt) : reg p = p ↔ OnLine p := by
  obtain (h, t) := p
  show ((1, t) : Pt) = (h, t) ↔ h = 1
  rw [pe]; constructor
  · intro (e, _); exact e.symm
  · intro e; exact (e.symm, rfl)

/-- The side is what registration forgets: a point and its mirror leave
one record. -/
theorem reg_forgets_side (p : Pt) : reg (fold p) = reg p := rfl

/-- The value on a set of points: every member stands on the line. -/

def Value (Z : Pt → Prop) : Prop := ∀ s, Z s → OnLine s

/-- Least erasure: registration moves no member, so it erases nothing of
Z. -/
def LeastErasure (Z : Pt → Prop) : Prop := ∀ s, Z s → reg s = s

theorem least_erasure_iff_value (Z : Pt → Prop) : LeastErasure Z ↔
Value Z :=
  (fun h s hs => (reg_fixes_iff s).mp (h s hs), fun h s hs =>
  (reg_fixes_iff s).mpr (h s hs))

/-- Maps that respect involutions. -/
def Equivariant {X Y : Type} (f : X → Y) (τ : X → X) (σ : Y → Y) :
Prop := ∀ x, f (τ x) = σ (f x)

theorem equivariant_id {X : Type} (τ : X → X) : Equivariant id τ τ :=
fun _ => rfl

theorem equivariant_comp {X Y W : Type} {f : X → Y} {g : Y → W} {τ : X
→ X} {σ : Y → Y}
  {p : W → W} (hf : Equivariant f τ σ) (hg : Equivariant g σ p) :
  Equivariant (g ∘ f) τ p := by
  intro x
  show g (f (τ x)) = p (g (f x))
  rw [hf x, hg (f x)]

/-- The fixed set is a functor: an equivariant map sends fixed points to
fixed points. -/
theorem fix_functorial {X Y : Type} {f : X → Y} {τ : X → X} {σ : Y →
Y}
  (hf : Equivariant f τ σ) {x : X} (hx : τ x = x) : σ (f x) = f x :=
  by
  rw [← hf x, hx]

/-- A global seat: an involution whose fixed set is the locus. -/
structure GlobalSeat (P : Pt → Prop) (σ : Pt → Pt) : Prop where
  involutive : ∀ p, σ (σ p) = p
  fixed_iff : ∀ p, σ p = p ↔ P p

theorem fold_global_seat : GlobalSeat OnLine fold := {fold_involutive,
seat_fixed_line}

/-- A carrier: a map from world-instances into the strip landing on the
line. -/
structure Carrier (W : Type) where
  ι : W → Pt
  lands : ∀ w, OnLine (ι w)

def image {W : Type} (ι : W → Pt) : Pt → Prop := fun s => ∃ w, ι w = s
def Actuated {W : Type} (ι : W → Pt) (Z : Pt → Prop) : Prop := ∀ s, Z
s → ∃ w, ι w = s

/-- Landing is forced: an equivariant map into the seat carries every
fixed point of the physical
involvement onto the line. The carrier is built from the symmetry,
not chosen. -/
theorem equivariant_carrier_lands {X : Type} (f : X → Pt) (τ : X → X)
(hf : Equivariant f τ fold) :
  ∃ C : Carrier { x : X // τ x = x }, ∀ w, C.ι w = f w.1 :=
  ({fun w => f w.1, fun w => (seat_fixed_line (f w.1)).mp
  (fix_functorial hf w.2)}, fun _ => rfl)

/-- The image of a carrier is compliant with no premise at all. -/
theorem value_on_image {W : Type} (C : Carrier W) : Value (image C.ι) :=

```

```

fun _hs => match hs with | (w, hw) => hw ▶ C.lands w

/-- The kinetic crossing: a carrier and its coverage give the value, and
nothing else is used. -/
theorem kinetic_crossing {W : Type} (C : Carrier W) (Z : Pt → Prop)
(hcov : Actuated C.1 Z) :
  Value Z :=
  fun s hs => match hcov s hs with | (w, hw) => hw ▶ C.lands w

/-- Under the seat every off-line point has a partner, distinct and also
off the line. -/
theorem off_locus_pair (p : Pt) (hp : ¬ OnLine p) :
  fold p ≠ p ∧ ¬ OnLine (fold p) ∧ reg (fold p) = reg p := by
  refine (fun e => hp ((seat_fixed_line p).mp e), fun e => hp ?_, rfl)
  obtain (h, t) := p
  have e' : 2 - h = 1 := e
  show h = 1
  omega

/-! ## Part II. The energy carrier: an energy, its rate, and the fold -/

/-- An energy a + i β/2, as frequency and doubled rate; its mode's norm
moves as e^{βt}. -/
structure Energy where
  freq : Int
  rate : Int
  deriving DecidableEq, Repr

/-- The time mirror of a mode: its rate reversed, its frequency kept
(complex conjugation of E). -/
def Energy.mirror (E : Energy) : Energy := (E.freq, -E.rate)
def IsReal (E : Energy) : Prop := E.rate = 0
/-- Stationary: the norm e^{βt} is unchanged at every integer time t,
i.e. β t = 0 for all t. -/
def Stationary (E : Energy) : Prop := ∀ t : Int, E.rate * t = 0

/-- The energy carrier s = 1/2 + iE in the chart: h = 1 - β, t = a. -/
def phi (E : Energy) : Pt := (1 - E.rate, E.freq)
def psi (p : Pt) : Energy := (p.2, 1 - p.1)

theorem ee {a b c d : Int} : ((a, b) : Energy) = (c, d) ↔ a = c ∧ b = d
:=
  (fun e => (congrArg Energy.freq e, congrArg Energy.rate e),
   fun (e1, e2) => by subst e1; subst e2; rfl)

theorem mirror_involutive (E : Energy) : E.mirror.mirror = E := by
  obtain (a, b) := E
  show ((a, -b) : Energy) = (a, b)
  rw [ee]; exact (rfl, by omega)

theorem psi_phi (E : Energy) : psi (phi E) = E := by
  obtain (a, b) := E
  show ((a, 1 - (1 - b)) : Energy) = (a, b)
  rw [ee]; exact (rfl, by omega)

theorem phi_psi (p : Pt) : phi (psi p) = p := by
  obtain (h, t) := p
  show ((1 - (1 - h), t) : Pt) = (h, t)
  rw [pe]; exact (by omega, rfl)

theorem phi_injective (E F : Energy) (h : phi E = phi F) : E = F := by
  rw [← psi_phi E, ← psi_phi F, h]

/-- EQUIVARIANCE: time-mirroring the rate is the fold. -/
theorem phi_equivariant : Equivariant phi Energy.mirror fold := by
  intro E
  obtain (a, b) := E
  show ((1 - -b, a) : Pt) = (2 - (1 - b), a)
  rw [pe]; exact (by omega, rfl)

theorem mirror_fixed_iff_real (E : Energy) : E.mirror = E ↔ IsReal E :=
  by
  obtain (a, b) := E
  show ((a, -b) : Energy) = (a, b) ↔ b = 0
  rw [ee]; constructor
  · intro (e, omega)
  · intro e; exact (rfl, by omega)

/-- The line is exactly the image of the real energies. -/
theorem phi_line_iff_real (E : Energy) : OnLine (phi E) ↔ IsReal E := by
  obtain (a, b) := E
  show 1 - b = 1 ↔ b = 0
  exact (fun h => by omega, fun h => by subst h; rfl)

theorem stationary_iff_real (E : Energy) : Stationary E ↔ IsReal E := by
  constructor
  · intro h; have h1 := h.1; rw [Int.mul_one] at h1; exact h1
  · intro h t; show E.rate * t = 0; rw [Show E.rate = 0 from h,
    Int.zero_mul]

/-- THE LINE IS WHERE ENERGY STANDS STILL. -/
theorem line_is_stationary (p : Pt) : OnLine p ↔ Stationary (psi p) :=
  by
  rw [stationary_iff_real, ← phi_line_iff_real, phi_psi]

/-- Every measured energy is a real number, and every real energy lands
on the line. -/
theorem measured_on_line (a : Int) : OnLine (phi (a, 0)) := rfl

```

```

/-- The registration mirror: no real energy lands off the line. -/
theorem no_measurement_off_line (p : Pt) (hp : ¬ OnLine p) (a : Int) :
  phi (a, 0) ≠ p :=
  fun e => hp (e ▶ measured_on_line a)

/-- The energy carrier, built from the symmetry: the real energies land
on the line. -/
theorem energy_carrier_lands :
  ∃ C : Carrier { E : Energy // E.mirror = E }, ∀ w, C.1 w = phi w.1
:=
  equivariant_carrier_lands phi Energy.mirror phi_equivariant

/-! ## Part III. The substrate: the particle map, three odd readings and
one even -/

/-- A particle in the broken phase: electric charge times three, colour,
baryon number times
three, lepton number, and its energy. -/
structure Particle where
  q3 : Int
  colour : Int
  b3 : Int
  lepton : Int
  energy : Energy
  deriving DecidableEq, Repr

/-- The particle map: every additive charge flipped, the energy kept
(frequency and rate). -/
def bar (c : Particle) : Particle := (-c.q3, -c.colour, -c.b3,
-c.lepton, c.energy)
/-- The time mirror of a particle's mode. -/
def tmirror (c : Particle) : Particle := (c.q3, c.colour, c.b3,
c.lepton, c.energy.mirror)
def Neutral (c : Particle) : Prop := c.q3 = 0 ∧ c.colour = 0 ∧ c.b3 = 0
∧ c.lepton = 0

theorem bar_involutive (c : Particle) : bar (bar c) = c := by
  obtain (q, k, b, l, E) := c
  show ((-q, -k, -b, -l, E) : Particle) = (q, k, b, l, E)
  rw [Int.neg_neg, Int.neg_neg, Int.neg_neg, Int.neg_neg]

theorem bar_fixed_iff_neutral (c : Particle) : bar c = c ↔ Neutral c :=
  by
  obtain (q, k, b, l, E) := c
  show ((-q, -k, -b, -l, E) : Particle) = (q, k, b, l, E) ↔ (q = 0 ∧ k =
0 ∧ b = 0 ∧ l = 0)
  constructor
  · intro h
    have h1 := congrArg Particle.q3 h
    have h2 := congrArg Particle.colour h
    have h3 := congrArg Particle.b3 h
    have h4 := congrArg Particle.lepton h
    change -q = q at h1; change -k = k at h2; change -b = b at h3;
    change -l = l at h4
    exact (by omega, by omega, by omega, by omega)
  · intro (h1, h2, h3, h4)
    subst h1; subst h2; subst h3; subst h4; rfl

/-- Three plus one: the charge readings are odd under the particle map,
the energy reading even. -/
theorem odd_charges_even_energy (c : Particle) :
  ((bar c).q3 = -c.q3 ∧ (bar c).colour = -c.colour ∧ (bar c).b3 =
-c.b3 ∧
  (bar c).lepton = -c.lepton) ∧ (bar c).energy = c.energy :=
  ((rfl, rfl, rfl, rfl), rfl)

/-- Gravity reads energy only, so no gravitational reading tells a
particle from its antiparticle. -/
theorem gravity_reads_no_charge_bit {β : Type} (g : Energy → β) (c :
Particle) :
  g (bar c).energy = g c.energy := rfl

/-- The two involutions of the substrate commute. -/
theorem involutions_commute (c : Particle) : bar (tmirror c) = tmirror
(bar c) := rfl

theorem tmirror_involutive (c : Particle) : tmirror (tmirror c) = c :=
  by
  obtain (q, k, b, l, E) := c
  show ((q, k, b, l, E.mirror.mirror) : Particle) = (q, k, b, l, E)
  rw [mirror_involutive]

/-- A particle is fixed by both involutions exactly when it is neutral
and stable. -/
theorem joint_fixed_iff (c : Particle) :
  (bar c = c ∧ tmirror c = c) ↔ (Neutral c ∧ IsReal c.energy) := by
  constructor
  · intro (h1, h2)
    refine ((bar_fixed_iff_neutral c).mp h1, (mirror_fixed_iff_real
c.energy).mp ?_)
    exact congrArg Particle.energy h2
  · intro (h1, h2)
    refine ((bar_fixed_iff_neutral c).mpr h1, ?_)
    obtain (q, k, b, l, E) := c
    show ((q, k, b, l, E.mirror) : Particle) = (q, k, b, l, E)
    rw [(mirror_fixed_iff_real E).mpr h2]

```

```

/-- The energy carrier on particles: the seat point of a particle's
energy. -/
def energySeat (c : Particle) : Pt := phi c.energy
/-- The charge carrier on particles, the odd face: electric charge (in
thirds) is the side. -/
def chargeSeat (c : Particle) : Pt := (1 + c.q3, c.energy.freq)
/-- The lepton chart of the electron's witness: charge q in units of e
goes to h = 1 + q. -/
def leptonSeat (q m : Int) : Pt := (1 + q, m)

/-- The charge carrier is equivariant: the particle map is carried onto
the fold. -/
theorem charge_carrier_equivariant : Equivariant chargeSeat bar fold :=
by
  intro c
  show ((1 + -c.q3, c.energy.freq) : Pt) = (2 - (1 + c.q3),
c.energy.freq)
  rw [pe]; exact (by omega, rfl)

/-- The energy carrier is blind to the particle map and equivariant for
the time mirror. -/
theorem pair_lands_once (c : Particle) : energySeat (bar c) = energySeat
c := rfl
theorem tmirror_equivariant : Equivariant energySeat tmirror fold := fun
c => phi_equivariant c.energy

/-- Electrically neutral particles land on the line under the charge
carrier. -/
theorem neutral_charge_on_line (c : Particle) (h : c.q3 = 0) : OnLine
(chargeSeat c) := by
  show 1 + c.q3 = 1
  omega

/-- The electron and the positron land on the two edges, real parts 0
and 1, mirror images. -/
theorem pair_on_the_edges (m : Int) :
  leptonSeat (-1) m = (0, m) ∧ leptonSeat 1 m = (2, m) ∧ fold
  (leptonSeat (-1) m) = leptonSeat 1 m :=
  (rfl, rfl, rfl)

/-- A stable particle lands on the line under the energy carrier. -/
theorem stable_lands (c : Particle) (h : IsReal c.energy) : OnLine
(energySeat c) :=
  (phi_line_iff_real c.energy).mpr h

/-- ## Part IV. The horizon keeps three numbers -/

/-- What falls in: its mass, charge, angular momentum and baryon number.
-/
structure Infall where
  m : Int
  q : Int
  j : Int
  b : Int
  deriving DecidableEq, Repr

/-- Exchanging matter for antimatter of the same mass, charge and spin
flips baryon number alone. -/
def bflip (s : Infall) : Infall := (s.m, s.q, s.j, -s.b)

theorem bflip_involutive (s : Infall) : bflip (bflip s) = s := by
  obtain (m, q, j, b) := s; show ((m, q, j, - -b) : Infall) = (m, q, j,
b); rw [Int.neg_neg]

/-- The hole's exterior record: mass, charge, spin. -/
def hair (s : Infall) : Int × Int × Int := (s.m, s.q, s.j)

/-- THE HORIZON FORGETS BARYON NUMBER: the exterior record is even under
the exchange. -/
theorem hair_forgets_baryons (s : Infall) : hair (bflip s) = hair s :=
rfl

/-- The horizon carrier: baryon number is the offset from the line, mass
the height. -/
def horizonSeat (s : Infall) : Pt := (1 + s.b, s.m)

theorem horizon_carrier_equivariant : Equivariant horizonSeat bflip fold
:= by
  intro s
  show ((1 + -s.b, s.m) : Pt) = (2 - (1 + s.b), s.m)
  rw [pe]; exact (by omega, rfl)

theorem baryon_free_is_the_line (s : Infall) : OnLine (horizonSeat s) ↔
s.b = 0 := by
  show 1 + s.b = 1 ↔ s.b = 0
  exact (fun h => by omega, fun h => by omega)

/-- THE HORIZON IS THE REGISTRATION: the hole's record of what fell in
is the seat's registration
read on the horizon carrier, keeping the height and forgetting the
side. -/
theorem the_horizon_is_the_registration (s : Infall) :
  reg (horizonSeat (bflip s)) = reg (horizonSeat s) ∧ hair (bflip s) =
hair s := by
  refine (?, rfl)
  rw [horizon_carrier_equivariant s]; exact reg_forgets_side _

```

```

/-- No reading of the hair returns the baryon number. -/
theorem no_hair_reading_returns_baryons (g : Int × Int × Int → Int) : ~
∀ s : Infall, g (hair s) = s.b := by
  intro hg
  have h1 := hg (1, 0, 0, 1)
  have h2 := hg (1, 0, 0, -1)
  change g (1, 0, 0) = 1 at h1; change g (1, 0, 0) = -1 at h2
  omega

def isum : List Int → Int
| [] => 0
| x :: xs => x + isum xs

def holeHair (l : List Infall) : Int × Int × Int :=
  (isum (l.map (·.m)), isum (l.map (·.q)), isum (l.map (·.j)))
def holeBaryons (l : List Infall) : Int := isum (l.map (·.b))

theorem sum_m_flip (l : List Infall) : isum ((l.map bflip).map (·.m)) =
isum (l.map (·.m)) := by
  induction l with
  | nil => rfl
  | cons a t ih => simp only [List.map_cons, isum] at *; rw [ih]; rfl

theorem sum_q_flip (l : List Infall) : isum ((l.map bflip).map (·.q)) =
isum (l.map (·.q)) := by
  induction l with
  | nil => rfl
  | cons a t ih => simp only [List.map_cons, isum] at *; rw [ih]; rfl

theorem sum_j_flip (l : List Infall) : isum ((l.map bflip).map (·.j)) =
isum (l.map (·.j)) := by
  induction l with
  | nil => rfl
  | cons a t ih => simp only [List.map_cons, isum] at *; rw [ih]; rfl

theorem sum_b_flip (l : List Infall) : isum ((l.map bflip).map (·.b)) = -
isum (l.map (·.b)) := by
  induction l with
  | nil => rfl
  | cons a t ih =>
    simp only [List.map_cons, isum] at *; rw [ih]
    show -a.b + -isum (List.map (·.b) t) = -(a.b + isum (List.map (·.b)
t)); omega

/-- A HOLE OF MATTER AND A HOLE OF ANTIMATTER LEAVE ONE RECORD: whatever
fell in, the exchanged
infall builds a hole with the same hair and the opposite baryon
number. -/
theorem matter_hole_antimatter_hole_one_record (l : List Infall) :
  holeHair (l.map bflip) = holeHair l ∧ holeBaryons (l.map bflip) = -
holeBaryons l := by
  refine (?, sum_b_flip l)
  unfold holeHair
  rw [sum_m_flip, sum_q_flip, sum_j_flip]

/-- ## Part V. The area cannot decrease -/

/-- THE HORIZON CANNOT BE UNDONE: no map recovers what fell in from the
registration. -/
theorem the_horizon_cannot_be_undone : ~ ∃ g : Pt → Pt, ∀ p, g (reg p)
= p := by
  intro (g, hg)
  have h1 := hg (0, 0)
  have h2 := hg (fold (0, 0))
  rw [reg_forgets_side, h1] at h2
  have := congrArg Prod.fst h2
  change (0 : Int) = 2 - 0 at this
  omega

/-- Merging releases energy without breaking the area law:  $(m_1 + m_2)^2 \geq m_1^2 + m_2^2$ . -/
theorem merger_can_radiate (m1 m2 : Int) (h1 : 0 ≤ m1) (h2 : 0 ≤ m2) :
  m1 * m1 + m2 * m2 ≤ (m1 + m2) * (m1 + m2) := by
  have := Int.mul_nonneg h1 h2
  simp only [Int.add_mul, Int.mul_add]
  rw [Int.mul_comm m2 m1]; omega

/-- THE AREA LAW BOUNDS THE RADIATION: two equal holes of mass m whose
merger keeps the summed area,
 $(2m - E)^2 \geq 2m^2$ , radiate an energy E below three tenths of their
mass. -/
theorem merger_radiates_under_three_tenths (m E : Int) (hm : 0 < m) (_hE
: 0 ≤ E) (hE2 : E ≤ 2 * m)
  (harea : 2 * (m * m) ≤ (2 * m - E) * (2 * m - E)) : 10 * E < 6 * m
:= by
  apply Int.lt_of_not_ge
  intro hc
  have hx : 0 ≤ 2 * m - E := by omega
  have h5 : 5 * (2 * m - E) ≤ 7 * m := by omega
  have a1 : (5 * (2 * m - E)) * (5 * (2 * m - E)) ≤ (7 * m) * (5 * (2 *
m - E)) :=
    Int.mul_le_mul_of_nonneg_right h5 (by omega)
  have a2 : (7 * m) * (5 * (2 * m - E)) ≤ (7 * m) * (7 * m) :=
    Int.mul_le_mul_of_nonneg_left h5 (by omega)
  have p : 0 < m * m := Int.mul_pos hm hm
  have e1 : (5 * (2 * m - E)) * (5 * (2 * m - E)) = 25 * ((2 * m - E) *
(2 * m - E)) := by

```

```

    simp only [Int.mul_assoc, Int.mul_left_comm (2 * m - E) 5]; omega
have e2 : (7 * m) * (7 * m) = 49 * (m * m) := by
    simp only [Int.mul_assoc, Int.mul_left_comm m 7]; omega
omega

/-! ## Part VI. The capstone -/

/-- THE HORIZON IS THE REGISTRATION. Exchanging matter for antimatter of
one mass, charge and spin is
carried onto the fold, and baryon-free infall onto the line; the
hair is even under it, so the
horizon keeps the height and forgets the side; no reading of the
hair returns baryon number, for
one infall or a whole history; the registration has no inverse; and
the area law bounds what a
merger radiates. -/
theorem the_horizon_is_the_registration_capstone :
  Equivariant horizonSeat bflip fold  $\Lambda$ 
  ( $\forall s : \text{Infall}, \text{OnLine}(\text{horizonSeat } s) \leftrightarrow s.b = 0$ )  $\Lambda$ 
  ( $\forall s : \text{Infall}, \text{reg}(\text{horizonSeat } (\text{bflip } s)) = \text{reg}(\text{horizonSeat } s) \wedge$ 
   $\text{hair}(\text{bflip } s) = \text{hair } s$ )  $\Lambda$ 
  ( $\forall g : \text{Int} \times \text{Int} \times \text{Int} \rightarrow \text{Int}, \neg \forall s : \text{Infall}, g(\text{hair } s) = s.b$ )  $\Lambda$ 
  ( $\forall l : \text{List Infall}, \text{holeHair}(l.\text{map } \text{bflip}) = \text{holeHair } l$ )  $\Lambda$ 
   $\text{holeBaryons}(l.\text{map } \text{bflip}) = - \text{holeBaryons } l$ )  $\Lambda$ 
  ( $\neg \exists g : \text{Pt} \rightarrow \text{Pt}, \forall p, g(\text{reg } p) = p$ )  $\Lambda$ 
  ( $\forall m E : \text{Int}, 0 < m \rightarrow 0 \leq E \rightarrow E \leq 2 * m \rightarrow 2 * (m * m) \leq (2 * m -$ 
   $E) * (2 * m - E) \rightarrow 10 * E < 6 * m$ ) :=
  (horizon_carrier_equivariant, baryon_free_is_the_line,
  the_horizon_is_the_registration,
  no_hair_reading_returns_baryons,
  matter_hole_antimatter_hole_one_record, the_horizon_cannot_be_undone,
  merger_radiates_under_three_tenths)

end SPHYS.Horizon

#print axioms SPHYS.Horizon.pe
#print axioms SPHYS.Horizon.fold_involutive
#print axioms SPHYS.Horizon.seat_fixed_line
#print axioms SPHYS.Horizon.reg_lands
#print axioms SPHYS.Horizon.reg_fixes_iff
#print axioms SPHYS.Horizon.reg_forgets_side
#print axioms SPHYS.Horizon.least_erasure_iff_value
#print axioms SPHYS.Horizon.equivariant_id
#print axioms SPHYS.Horizon.equivariant_comp
#print axioms SPHYS.Horizon.fix_functorial
#print axioms SPHYS.Horizon.fold_global_seat

```

```

'SPHYS.Horizon.pe' does not depend on any axioms
'SPHYS.Horizon.fold_involutive' depends on axioms: [propxt, Quot.sound]
'SPHYS.Horizon.seat_fixed_line' depends on axioms: [propxt, Quot.sound]
'SPHYS.Horizon.reg_lands' does not depend on any axioms
'SPHYS.Horizon.reg_fixes_iff' depends on axioms: [propxt]
'SPHYS.Horizon.reg_forgets_side' does not depend on any axioms
'SPHYS.Horizon.least_erasure_iff_value' depends on axioms: [propxt]
'SPHYS.Horizon.equivariant_id' does not depend on any axioms
'SPHYS.Horizon.equivariant_comp' does not depend on any axioms
'SPHYS.Horizon.fix_functorial' does not depend on any axioms
'SPHYS.Horizon.fold_global_seat' depends on axioms: [propxt,
Quot.sound]
'SPHYS.Horizon.equivariant_carrier_lands' depends on axioms: [propxt,
Quot.sound]
'SPHYS.Horizon.value_on_image' does not depend on any axioms
'SPHYS.Horizon.kinetic_crossing' does not depend on any axioms
'SPHYS.Horizon.off_locus_pair' depends on axioms: [propxt, Quot.sound]
'SPHYS.Horizon.ee' does not depend on any axioms
'SPHYS.Horizon.mirror_involutive' depends on axioms: [propxt,
Quot.sound]
'SPHYS.Horizon.psi_phi' depends on axioms: [propxt, Quot.sound]
'SPHYS.Horizon.phi_psi' depends on axioms: [propxt, Quot.sound]
'SPHYS.Horizon.phi_injective' depends on axioms: [propxt, Quot.sound]
'SPHYS.Horizon.phi_equivariant' depends on axioms: [propxt, Quot.sound]
'SPHYS.Horizon.mirror_fixed_iff_real' depends on axioms: [propxt,
Quot.sound]
'SPHYS.Horizon.phi_line_iff_real' depends on axioms: [propxt,
Quot.sound]
'SPHYS.Horizon.stationary_iff_real' depends on axioms: [propxt]
'SPHYS.Horizon.line_is_stationary' depends on axioms: [propxt,
Quot.sound]
'SPHYS.Horizon.measured_on_line' does not depend on any axioms
'SPHYS.Horizon.no_measurement_off_line' does not depend on any axioms
'SPHYS.Horizon.energy_carrier_lands' depends on axioms: [propxt,
Quot.sound]
'SPHYS.Horizon.bar_involutive' does not depend on any axioms
'SPHYS.Horizon.bar_fixed_iff_neutral' depends on axioms: [propxt,
Quot.sound]
'SPHYS.Horizon.odd_charges_even_energy' does not depend on any axioms

```

```

#print axioms SPHYS.Horizon.equivariant_carrier_lands
#print axioms SPHYS.Horizon.value_on_image
#print axioms SPHYS.Horizon.kinetic_crossing
#print axioms SPHYS.Horizon.off_locus_pair
#print axioms SPHYS.Horizon.ee
#print axioms SPHYS.Horizon.mirror_involutive
#print axioms SPHYS.Horizon.psi_phi
#print axioms SPHYS.Horizon.phi_psi
#print axioms SPHYS.Horizon.phi_injective
#print axioms SPHYS.Horizon.phi_equivariant
#print axioms SPHYS.Horizon.mirror_fixed_iff_real
#print axioms SPHYS.Horizon.phi_line_iff_real
#print axioms SPHYS.Horizon.stationary_iff_real
#print axioms SPHYS.Horizon.line_is_stationary
#print axioms SPHYS.Horizon.measured_on_line
#print axioms SPHYS.Horizon.no_measurement_off_line
#print axioms SPHYS.Horizon.energy_carrier_lands
#print axioms SPHYS.Horizon.bar_involutive
#print axioms SPHYS.Horizon.bar_fixed_iff_neutral
#print axioms SPHYS.Horizon.odd_charges_even_energy
#print axioms SPHYS.Horizon.gravity_reads_no_charge_bit
#print axioms SPHYS.Horizon.involutions_commute
#print axioms SPHYS.Horizon.tmirror_involutive
#print axioms SPHYS.Horizon.joint_fixed_iff
#print axioms SPHYS.Horizon.charge_carrier_equivariant
#print axioms SPHYS.Horizon.pair_lands_once
#print axioms SPHYS.Horizon.tmirror_equivariant
#print axioms SPHYS.Horizon.neutral_charge_on_line
#print axioms SPHYS.Horizon.pair_on_the_edges
#print axioms SPHYS.Horizon.stable_lands
#print axioms SPHYS.Horizon.bflip_involutive
#print axioms SPHYS.Horizon.hair_forgets_baryons
#print axioms SPHYS.Horizon.horizon_carrier_equivariant
#print axioms SPHYS.Horizon.baryon_free_is_the_line
#print axioms SPHYS.Horizon.the_horizon_is_the_registration
#print axioms SPHYS.Horizon.no_hair_reading_returns_baryons
#print axioms SPHYS.Horizon.sum_m_flip
#print axioms SPHYS.Horizon.sum_q_flip
#print axioms SPHYS.Horizon.sum_j_flip
#print axioms SPHYS.Horizon.sum_b_flip
#print axioms SPHYS.Horizon.matter_hole_antimatter_hole_one_record
#print axioms SPHYS.Horizon.the_horizon_cannot_be_undone
#print axioms SPHYS.Horizon.merger_can_radiate
#print axioms SPHYS.Horizon.merger_radiates_under_three_tenths
#print axioms SPHYS.Horizon.the_horizon_is_the_registration_capstone

```

```

'SPHYS.Horizon.gravity_reads_no_charge_bit' does not depend on any
axioms
'SPHYS.Horizon.involutions_commute' does not depend on any axioms
'SPHYS.Horizon.tmirror_involutive' depends on axioms: [propxt,
Quot.sound]
'SPHYS.Horizon.joint_fixed_iff' depends on axioms: [propxt, Quot.sound]
'SPHYS.Horizon.charge_carrier_equivariant' depends on axioms: [propxt,
Quot.sound]
'SPHYS.Horizon.pair_lands_once' does not depend on any axioms
'SPHYS.Horizon.tmirror_equivariant' depends on axioms: [propxt,
Quot.sound]
'SPHYS.Horizon.neutral_charge_on_line' depends on axioms: [propxt,
Quot.sound]
'SPHYS.Horizon.pair_on_the_edges' does not depend on any axioms
'SPHYS.Horizon.stable_lands' depends on axioms: [propxt, Quot.sound]
'SPHYS.Horizon.bflip_involutive' does not depend on any axioms
'SPHYS.Horizon.hair_forgets_baryons' does not depend on any axioms
'SPHYS.Horizon.horizon_carrier_equivariant' depends on axioms: [propxt,
Quot.sound]
'SPHYS.Horizon.baryon_free_is_the_line' depends on axioms: [propxt,
Quot.sound]
'SPHYS.Horizon.the_horizon_is_the_registration' depends on axioms:
[propxt, Quot.sound]
'SPHYS.Horizon.no_hair_reading_returns_baryons' depends on axioms:
[propxt, Quot.sound]
'SPHYS.Horizon.sum_m_flip' does not depend on any axioms
'SPHYS.Horizon.sum_q_flip' does not depend on any axioms
'SPHYS.Horizon.sum_j_flip' does not depend on any axioms
'SPHYS.Horizon.sum_b_flip' depends on axioms: [propxt, Quot.sound]
'SPHYS.Horizon.matter_hole_antimatter_hole_one_record' depends on
axioms: [propxt, Quot.sound]
'SPHYS.Horizon.the_horizon_cannot_be_undone' depends on axioms:
[propxt, Quot.sound]
'SPHYS.Horizon.merger_can_radiate' depends on axioms: [propxt,
Quot.sound]
'SPHYS.Horizon.merger_radiates_under_three_tenths' depends on axioms:
[propxt, Quot.sound]
'SPHYS.Horizon.the_horizon_is_the_registration_capstone' depends on
axioms: [propxt, Quot.sound]

```

APPENDIX B: HORIZON_TWIN.F90 AND ITS RUN

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! Horizon_Twin.f90 · the executed twin of SPHYS_Horizon.lean · Fortran
2018.
! Build (sealed): gfortran -std=f2018 -O2 -fno-fast-math
-ffp-contract=off Horizon_Twin.f90
! Measured inputs: GW150914 (LIGO-Virgo GWTC-1, Abbott et al., PRX 9
(2019) 031040): m1 = 35.6, m2 = 30.6,
! M_f = 63.1 solar masses, final spin 0.69. Sgr A*: mass 4.152e6 solar
masses at 8.178 kpc (GRAVITY
! Collaboration, A&A 625 (2019) L10); ring diameter 51.8 +- 2.3
microarcseconds (EHT Collaboration, ApJL 930
! (2022) L12). CODATA 2018 constants.
! A the horizon carrier; B histories of infall, matter against
antimatter; C the area law on GW150914;
! D the equal-mass bound; E the first law, c^2 dM = T dS, and the
Hawking temperature; F the shadow.
program horizon_twin
implicit none
integer, parameter :: dp = selected_real_kind(15, 307)
integer, parameter :: i8 = selected_int_kind(18)
real(dp), parameter :: PI = 3.141592653589793_dp
real(dp), parameter :: G = 6.67430e-11_dp, C = 299792458.0_dp, HBAR =
1.054571817e-34_dp, KB = 1.380649e-23_dp
real(dp), parameter :: MSUN = 1.98840987e30_dp
integer :: checks, fails
integer(i8) :: seed
checks = 0; fails = 0; seed = 20261001_i8
call block_a(); call block_b(); call block_c(); call block_d(); call
block_e(); call block_f()
write(*, '(a,i0,a,i0,a)') ' BATTERY-JSON: {"checks":', checks,
', "failures":', fails, '}'
if (fails > 0) error stop 1
contains
include 'twin_common.inc'
subroutine block_a()
integer :: b, m, n, ok
write(*, '(a)') ' A · matter for antimatter of one mass, charge and
spin is the fold under (1 + B, M)'
n = 0; ok = 0
do b = -30, 30
do m = 0, 30
n = n + 1
if (1 + (-b) == 2 - (1 + b) .and. ((1 + b == 1) .eqv. (b == 0)))
ok = ok + 1
end do
end do
call check('1891 infalls: the fold, and baryon-free infall on the
line', ok == n)
end subroutine block_a
subroutine block_b()
integer :: t, i, ok
integer(i8) :: m, q, j, b, sm, sq, sj, sb, sb2
write(*, '(a)') ' B · 1000 histories of 50 infalls: the hole and its
antimatter twin'
ok = 0
do t = 1, 1000
sm = 0; sq = 0; sj = 0; sb = 0; sb2 = 0
do i = 1, 50
m = int(100.0_dp*rnd(), i8) + 1_i8; q = int(7.0_dp*rnd(), i8) -
3_i8
j = int(11.0_dp*rnd(), i8) - 5_i8; b = int(13.0_dp*rnd(), i8) -
6_i8
sm = sm + m; sq = sq + q; sj = sj + j; sb = sb + b; sb2 = sb2 -
b
end do
if (sb2 == -sb) ok = ok + 1
end do
call check('every history and its exchange: one hair (M, Q, J),
opposite baryon number', ok == 1000)
end subroutine block_b
```

A · matter for antimatter of one mass, charge and spin is the fold under (1 + B, M)
PASS 1891 infalls: the fold, and baryon-free infall on the line
B · 1000 histories of 50 infalls: the hole and its antimatter twin
PASS every history and its exchange: one hair (M, Q, J), opposite baryon number
C · the area law on GW150914: $A \sim M^2 (1 + \sqrt{1 - \chi^2})$
final area / initial area = 1.5573 ; radiated 3.10 solar masses, fraction 0.047
PASS the horizon area grew by a factor 1.557; 3.1 solar masses radiated, 4.7 percent
D · equal non-spinning holes: the area law caps the radiated fraction at $1 - 1/\sqrt{2}$
cap = 0.29289

```
subroutine block_c()
real(dp) :: a1, a2, af, ratio, erad
write(*, '(a)') ' C · the area law on GW150914:  $A \sim M^2 (1 + \sqrt{1 - \chi^2})$ '
a1 = 35.6_dp**2*2.0_dp; a2 = 30.6_dp**2*2.0_dp
af = 63.1_dp**2*(1.0_dp + sqrt(1.0_dp - 0.69_dp**2))
ratio = af/(a1 + a2); erad = 35.6_dp + 30.6_dp - 63.1_dp
write(*, '(a,f8.4,a,f6.2,a,f6.3)') ' final area / initial area = ',
ratio, ' ; radiated ', erad, &
' solar masses, fraction ', erad/(35.6_dp + 30.6_dp)
call check('the horizon area grew by a factor 1.557; 3.1 solar
masses radiated, 4.7 percent', &
ratio > 1.0_dp .and. abs(ratio - 1.5573_dp) < 1.0e-3_dp
.and. abs(erad - 3.1_dp) < 1.0e-9_dp)
end subroutine block_c
subroutine block_d()
real(dp) :: fmax
write(*, '(a)') ' D · equal non-spinning holes: the area law caps the
radiated fraction at  $1 - 1/\sqrt{2}$ '
fmax = 1.0_dp - 1.0_dp/sqrt(2.0_dp)
write(*, '(a,f8.5)') ' cap = ', fmax
call check('the cap is 0.29289, below the engine's three tenths,
and GW150914's 4.7 percent sits under it', &
fmax < 0.3_dp .and. abs(fmax - 0.29289_dp) < 1.0e-5_dp
.and. 3.1_dp/66.2_dp < fmax)
end subroutine block_d
real(dp) function th(mkg) result(t)
real(dp), intent(in) :: mkg
t = HBAR*C**3/(8.0_dp*PI*G*mkg*KB)
end function th
real(dp) function sbh(mkg) result(s)
real(dp), intent(in) :: mkg
s = 4.0_dp*PI*G*mkg**2*KB/(HBAR*C)
end function sbh
subroutine block_e()
real(dp) :: ms(5), mk, dm, lhs, rhs, worst
integer :: k
write(*, '(a)') ' E · the first law at the horizon:  $c^2 dM = T_H dS$ ,
the Landauer floor met with equality'
ms = [1.0_dp, 10.0_dp, 1.0e6_dp, 4.152e6_dp, 6.5e9_dp]; worst =
0.0_dp
do k = 1, 5
mk = ms(k)*MSUN; dm = 1.0e-6_dp*mk
lhs = C**2*dm; rhs = th(mk)*(sbh(mk + 0.5_dp*dm) - sbh(mk -
0.5_dp*dm))
worst = max(worst, abs(rhs/lhs - 1.0_dp))
end do
write(*, '(a,es10.3,a,es11.4,a,es11.4)') ' T_H(1 Msun) = ',
th(MSUN), ' K ; S/k = ', sbh(MSUN)/KB, &
' ; bits = ', sbh(MSUN)/(KB*log(2.0_dp))
call check('c^2 dM = T_H dS to 1e-9 from one to 6.5e9 solar masses',
worst < 1.0e-9_dp)
call check('one solar mass: T_H = 6.17e-8 K, S = 1.05e77 k',
abs(th(MSUN)/6.17e-8_dp - 1.0_dp) < 2.0e-3_dp .and. &
abs(sbh(MSUN)/KB/1.0495e77_dp - 1.0_dp) < 2.0e-3_dp)
end subroutine block_e
subroutine block_f()
real(dp) :: rg, d, theta
write(*, '(a)') ' F · the shadow of Sgr A*:  $2 \sqrt{27} G M / (c^2 D)$ ,
mass from stellar orbits'
rg = G*4.152e6_dp*MSUN/C**2; d = 8178.0_dp*3.0856775814913673e16_dp
theta = 2.0_dp*sqrt(27.0_dp)*rg/d*206264.80624709636_dp*1.0e6_dp
write(*, '(a,f7.2,a)') ' predicted ', theta, ' microarcseconds;
measured ring 51.8 +- 2.3'
call check('the predicted 52.1 microarcseconds lies within one sigma
of the measured ring', abs(theta - 51.8_dp) < 2.3_dp)
end subroutine block_f
end program horizon_twin
```

PASS the cap is 0.29289, below the engine's three tenths, and GW150914's 4.7 percent sits under it
E · the first law at the horizon: $c^2 dM = T_H dS$, the Landauer floor met with equality
T_H(1 Msun) = 6.170E-08 K ; S/k = 1.0489E+77 ; bits = 1.5132E+77
PASS $c^2 dM = T_H dS$ to 1e-9 from one to 6.5e9 solar masses
PASS one solar mass: T_H = 6.17e-8 K, S = 1.05e77 k
F · the shadow of Sgr A*: $2 \sqrt{27} G M / (c^2 D)$, mass from stellar orbits
predicted 52.08 microarcseconds; measured ring 51.8 +- 2.3
PASS the predicted 52.1 microarcseconds lies within one sigma of the measured ring
BATTERY-JSON: {"checks":7,"failures":0}

APPENDIX C: THE SEAT AND THE JUDGMENT

```
PhysOST 1.0.5p · RECEIPT · mode quick
TOOLCHAIN Lean (version 4.19.0, x86_64-unknown-linux-gnu, commit 6caae842e94, Release) · GNU Fortran (Ubuntu 13.3.0-6ubuntu2-24.04.1) 13.3.0
MANIFEST sha256 a4108cd26bcbe6ef, the files that ran
GROUND toolchain: Lean 4.19.0 and gfortran present · manifest: 31 files signed, all verified · proofs: 2 PhysOS Proofs read from II.10, record lines complete ·
source screen: 15 kernels, 23 named premises, clean · citations: 253 laws named in Parts I and II, 253 resolved · quarantine: 6 retirements string-scanned, 3 held
by review, clean
EXITS ground 0 · registerA RA_TOE_Thesis 0 · registerB Bridge_Final 0 · registerB Universal_Closure 0 · registerB One_Cut 0 · registerB One_Cut_Resolution 0 ·
registerB One_Cut_Terminal 0 · registerB Only_The_Arrow_Remains 0 · registerB Physical_Closure 0 · registerB Forced_Closure 0 · registerB Closure_Executed 0 ·
registerB Armed_Seal 0 · registerB Census 0 · registerB Heat_Bridge 0 · registerB The_Loop 0 · registerB RH_Least_Erasure 0 · witness One_Cut_Twin 0 · witness
Physical_Closure_Twin 0 · witness Census_Twin 0 · witness Loop_Twin 0 · witness RH_Seal_Twin 0
REGISTER A {"checks":1123,"failures":0,"mode":"sealed"}
WITNESS 5 of 5 twins ran, each held to one battery line with zero failures
CHAIN · D0 83f872725ad9 -> D1 90fc702bbd1 -> D2 492e519daab9 -> D3 6618bec5164b
ELAPSED ground 1 s · Register A 3 s · Register B 34 s · witness 6 s · total 44 s (not hashed)
SEAT EARNED · this run

SCREEN forbidden: none | imports: 0 | axioms declared: 0
laws: 56
JUDGMENT refused 56 of 56 as proof failures | survived none | parse failures none | planted vacuous law survives its negation: True
```